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India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Session: Data Governance Framework and Data Management in Utilities

Building Data-Driven Digital Utilities

Presented By

Reji Kumar Pillai, President, India Smart Grid Forum

Why Data Matters in Modern Utilities

India SMART UTILITY Week 2026 | ISGF India Smart Grid Forum

Electric utilities are shifting from infrastructure-centric to data-centric operations

Explosion of Data Sources

- Smart meters (every 15–30 minutes)
- SCADA / EMS / DMS / PMU (milliseconds)
- Rooftop solar, EV chargers, batteries
- IoT sensors, drones, field devices
- Customer and billing systems

👉 Large utilities generate **terabytes of data daily**

Data is Now a Strategic Asset

- Data replaces guesswork
- Algorithms augment human operators
- Digital twins mirror physical grids
- AI becomes the "nervous system" of the grid

AIDA Annual Publication 2026: Data Management and Data Governance in Electric Utilities: Key Practices and Emerging Trends

Authors: Reji Kumar Pillai, President, ISGF and Shashi Bala, Deputy General Manager, PowerXchange Innovations

A. INTRODUCTION

Electric utilities today are undergoing one of the biggest transformations in their history. For more than a century, power utilities were handling wires, poles, transformers, power plants, substations and control rooms. Decisions were made based on human intuition, manual records, and limited supervisory data. Today, data has become the backbone of every modern electricity system. Meter data from smart meters in every 15-30 minutes, grid health from SCADA/EMS/DMS/PMU systems in milliseconds from thousands of nodes; and rooftop solar panels, EV chargers, and home batteries constantly send data to utility platforms. Drones, IoT sensors, and field devices generate humungous amounts of operational data. Today, a large utility generates several terabytes of new data every day.

In this new digital era, the utilities that succeed are the ones that treat data not as a by- product—but as a strategic asset as:

- *Data replaces guesswork*
- *Algorithms augment operators*
- *Digital twins complement physical networks*
- *AI becomes the nervous system of the grid*

This Paper attempts to articulate how modern utilities are managing and governing their data, the frameworks they use, the maturity levels in data management, and the roadmap that can help electric utilities in India to evolve into data-driven digital utilities.

control room is constantly communicating with each through data. If the data is bad or incomplete it can impact grid operations seriously.

Good Data Leads to	Poor Data Leads to
• Error free bills • Low technical and commercial losses • Better load and generation forecasting • Better renewable energy integration • Lesser outages and faster outage restoration • Better regulatory submissions • Better customer experience • Better network optimisation and efficient grid operation	• Wrong bills • Missing meter reads • High technical and commercial losses • Unreliable information to consumers and regulators • Delayed outage restoration • Unhappy customers • Poor network visibility and inefficient grid operation

This is why data governance and data management have become core functions—just like maintenance, billing, and grid operations.

B. DIGITAL JOURNEY OF INDIAN UTILITIES

Electric utilities India began their digital transformation with the R-APDRP program in 2009, which introduced:

https://aida-india.org/wp-content/uploads/2026/01/aida-annual-publication-2025-17-01-2026-final-for-print.pdf

Impact of Data Quality on Utility Performance

✓ Good Data Enables	❌ Poor Data Leads To
Accurate billing	Wrong bills and disputes
Loss reduction (technical & commercial)	High losses
Reliable forecasting	Poor demand planning
Renewable integration	Difficulty managing solar/wind variability
Faster outage restoration	Delayed restoration
Better regulatory reporting	Regulatory risks & penalties
Improved customer experience	Customer dissatisfaction
Efficient grid operation	Poor grid visibility
Complete meter readings	Missing or incorrect meter reads

👉 **Data governance is now as critical as operations and maintenance in modern utilities**

Digital Journey of Indian Utilities

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Phase	Focus	Key Outcomes
R-APDRP (2009)	IT Foundations	➢ GIS mapping ➢ Feeder metering ➢ Billing systems ➢ On premise Data centres and DR
Smart Grid Initiatives	Operational Automation	➢ SCADA/DMS ➢ OMS ➢ AMI pilot projects ➢ DR pilots
RDSS Era (Present)	Data-Driven Utilities	➢ ~250M smart meters ➢ Real-time data at scale ➢ Data lakes on the cloud ➢ Analytics-led operations

⚠ Key Challenges Today

- Fragmented data
- Interoperability gaps
- Poor data quality
- Missing metadata
- IT-OT silos
- Unclear data sharing

Data Management and Governance Framework

Data Management Lifecycle (End-to-End)
Collect → Clean → Organize → Store → Use
Sources: Smart meters • SCADA • GIS • Billing • DERs
Process: Error removal • **Structured hierarchy** (Substation → Feeder → DT → Consumer)
Platform: Secure cloud / data lakes
Outcome: Analytics • Forecasting • Operational decisions

👉 **Key Objective: Single Source of Truth**

Data Governance — "Traffic Rules for Data"
Ensures data is:

- Accurate and reliable
- Secure and protected
- Accessible to authorized users
- Auditable and traceable
- Compliant with regulations

Federated Data Governance Framework in Modern Utilities

Modern utilities adopt a **Federated Data Governance Model** that combines **central strategic control with domain-level data ownership**, ensuring accountability, data quality, and efficient decision-making across IT and OT systems

Governance Structure

- **Enterprise Data Governance Committee**

Led by the **Chief Digital Officer (CDO)** to define **data strategy, approve investments, resolve conflicts, and ensure cross-department alignment**

- **Data Owners (Domain Leaders)**

Business heads responsible for data in domains such as **Grid Operations, Customer Services, AMI, Billing, DER, Asset Management, and Regulatory Reporting**

- **Data Stewards (Operational Custodians)**

Teams responsible for **data quality, metadata accuracy, daily monitoring, and issue resolution**

- **Data Centre of Excellence (Data CoE)**

IT-led unit responsible for **data architecture, tools selection, data pipelines, data lake management, and advanced analytics development**

Outcome: **Trusted data, clear ownership, and faster data-driven decisions across the utility**

Future Architecture and Roadmap

How Utilities should Handle Data

Raw Data → Clean Data → Useful Information → Apps & Decisions

- Data comes from smart meters, sensors, grid systems
- It is cleaned and organized
- Converted into meaningful reports
- Used in dashboards, AI systems, and mobile apps

👉 **Like turning raw ingredients into ready-to-eat food**

What Will Become Important

- Managing rooftop solar, batteries, EVs
- Strong cybersecurity
- Safe sharing of data with partners
- Responsible use of AI
- Handling data generated near homes ("grid edge")

Step-by-Step Plan for DISCOMs

Short Term: Fix policies, roles, and messy data
Mid Term: Build cloud systems and analytics
Long Term: Fully smart, AI-driven power utility

Time Horizon	Focus	Key Actions
Short Term (1–2 yrs)	Build Foundation	➢ Policies ➢ Governance roles ➢ Data cleanup ➢ Initial dashboards
Mid Term (2–3 yrs)	Strengthen Core	➢ Cloud data lake ➢ System integration ➢ Metadata and security ➢ Analytics pilots
Long Term (3–5 yrs)	Digital Utility	➢ AI-driven operations ➢ DER/EV integration ➢ Automation ➢ Consumer-centric services

Key Message

👉 **Future utilities will run on data and intelligence — not just electricity supply**

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THANK YOU

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Links/References (If any)